

Research Idea: ngc_3256

ALMA Thesis Planner (automated pipeline)

Dense-gas excitation in the high-velocity wings versus the systemic core of NGC 3256: a spectral-decomposition HCN/HCO+ ladder study using archival ALMA data (project 2017.1.00379.S)

Using only the public, pipeline-calibrated products of ALMA project 2017.1.00379.S, the student will measure whether the dense molecular gas at high velocities in NGC 3256 — the component plausibly entrained in the merger’s known molecular outflow — is excited differently from the gas at systemic velocities in the central starburst. Because the two nuclei are only $\sim 5''$ apart and the coarsest beam in the set (Band 5, $\sim 7''$) blends them with the outflow base, velocity, not position, is the axis along which this dataset can genuinely separate the components, and the entire design is built around that fact. The identification of the high-velocity wings with the outflow is treated throughout as a physically motivated working hypothesis, tested internally where the data allow, never as an assumed result.

Plan of work:

1. **Data (fixed scope).** Download the QA2 image cubes for three member OUSs: Band 5 ($\sim 7''$), the low-resolution Band 6 ($\sim 4.4''$), and Band 7 ($\sim 3.2''$). The $0.24''$ and $0.8''$ Band 6 cubes are excluded from the science analysis except for two contained diagnostic uses (steps 5 and 6). No re-imaging of measurement sets unless a QA2 cube is demonstrably unusable, and any such exception is documented.

2. **Go/no-go gate (weeks 1–2), three checks.** First, verify that the Band 5 cube usably covers HCN(2–1) and HCO+(2–1): both lines fall near window edges (~ 175.6 and ~ 176.7 GHz observed), so confirm the full expected line-plus-wing velocity range (at least ± 400 km/s about systemic) lies within valid channels before any science is attempted. Second, inspect continuum-subtraction quality in all three QA2 cubes near the bright nuclei, since a poor baseline can mimic or mask faint line wings; any residual continuum is corrected with a simple image-plane baseline fit before proceeding. Third, convolve all cubes to the $\sim 7''$ Band 5 beam, extract a spectrum from one fixed circular aperture centered between the nuclei and enclosing both plus the outflow base, and measure the integrated signal-to-noise of the high-velocity wings ($|v - v_{\text{sys}}| \geq 200$ km/s) in the J=2–1 lines. If the wings are detected at 3σ , proceed with the full three-transition ladder. If the coverage or detection check fails, the thesis pivots to a two-transition (3–2, 4–3) wing/core study using Bands 6 and 7 only, with the Band 5 result entering as a quantified J=2–1 upper limit (or as excluded coverage, stated as such). The pivot is a complete, defensible thesis but is explicitly the weaker science case: with a single ladder ratio per molecule, the density–temperature degeneracy in the non-LTE grids widens substantially, and the fallback’s conclusions will be framed as bounds plus HCN/HCO+ chemistry contrasts rather than tight excitation constraints.

3. **Spectral decomposition.** For each of HCN and HCO+ at each available J, decompose the aperture spectrum into a systemic core and blue/red wing components by two independent methods:

(a) fixed velocity-window integration, with core and wing windows defined once from the brightest line and applied identically to every line — this is the primary measurement; (b) multi-Gaussian fitting with velocity centroids shared across transitions, as the robustness check.

4. Excitation and chemistry diagnostics. For core and wings separately, compute the HCN and HCO+ ladder ratios (3–2/2–1 and 4–3/2–1, or 4–3/3–2 alone under the pivot) and the HCN/HCO+ ratio at matched J. Interpret them against a pre-computed RADEX grid in density and kinetic temperature to bracket the physical conditions of the wing gas relative to the core gas. H¹³CN(4–3) in the Band 7 coverage provides an opacity check on the core component; for the wings it will almost certainly yield only an optical-depth limit, reported as such. Per-band flux-calibration uncertainty is propagated explicitly and is expected to dominate the cross-band ratios.

5. Flux-recovery systematic (bounded, using the dataset’s own redundancy). The same Band 6 lines are observed at ~0.24”, ~0.8”, and ~4.4”: the student compares the common-aperture flux between the ~0.8” and ~4.4” Band 6 cubes to measure how much flux the higher-resolution configuration resolves out, and adopts that number as the systematic-error *floor* on all cross-band ratios. This comparison is a proxy, not a direct calibration: it characterizes flux recovery within Band 6 only, while the science ratios also cross Band 5 and Band 7 cubes with different uv coverage and maximum recoverable scales. The thesis will state explicitly that the true cross-band systematic could exceed this floor, and every conclusion will be stress-tested under an inflated systematic before being claimed; in the worst case the headline result honestly degrades from a quantitative excitation bracket to a robust sign statement (“the wing gas is more/less excited than the core gas”), which is retained as an acceptable thesis outcome.

6. Wing-origin check (hard-capped at two weeks). In a violent merger, gas at $|v - v_{\text{sys}}|$ 200 km/s inside an aperture enclosing two nuclei can include nuclear rotation and tidal streaming, not only outflow. Using the ~3–4” Band 6/7 cubes (not the excluded highest resolutions), the student maps the centroid position of the wing emission and tests whether it is displaced from the nuclei roughly along the minor axis, as expected for outflowing gas, or instead tracks the nuclear rotation pattern. This check is time-boxed to two weeks and an ambiguous outcome is an acceptable and anticipated result: whichever way it comes out — including inconclusive — it is reported, and the interpretation section carries “wings = outflow” as an explicit caveat whose weight depends on this check. No external data are imported to settle it.

7. Write-up. The bounded central question — is the high-velocity dense gas more or less excited than the systemic nuclear gas, and what does that imply for the outflow’s ability to form stars? — is answered with a fixed set of quantified ratios, RADEX-bracketed conditions, honestly propagated statistical and systematic uncertainties, and clearly labeled interpretation caveats. Either gate outcome yields a complete thesis, with the relative strength of the two outcomes stated candidly rather than equated. The seven-step plan as written is the full scope; no additional analyses are added during supervision, because the design is deliberately at the complexity ceiling appropriate for one student.

Why this is thesis-scoped: it uses one archival project’s delivered products only; the realistic early failure modes (edge-of-window line coverage, imperfect continuum subtraction near the bright nuclei, and wing non-detection in the weakest band) are all confronted in the first two weeks with a pre-committed pivot rather than discovered in month six; the dominant interferometric systematic is bounded with data the project already contains instead of requiring new observations; the hardest interpretive assumption is tested internally within a hard two-week cap; and the deliverable is a small, fixed set of numbers — core and wing ratios for two molecules across two to three transitions

with a lookup-grid interpretation — rather than open-ended cloud-scale modeling.